

REVIEW ARTICLE

Education level and breast cancer incidence: a meta-analysis of cohort studies

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Abstract

Objective: We aimed to examine the association between education level and breast cancer incidence by a meta-analysis of cohort studies.

Methods: Relevant studies were identified by searching PubMed, Web of Science, and Scopus databases on June 10, 2018. Reference lists from the obtained articles were also reviewed. We included cohort studies reporting relative risks with 95% CIs for the association between education level and breast cancer incidence. Either a fixed- or random-effects model was used to calculate the pooled risk estimates.

Results: We identified 18 cohort studies with more than 10 million women. Compared with women with a lower education level, women with a higher education level had a significantly higher risk of developing breast cancer (pooled relative risk 1.22 [95% CI, 1.14-1.30]). The results did not differ by study area or reference group. The association remained in studies that adjusted for age at first birth and parity, but was attenuated and no longer significant when the analysis was restricted to studies that adjusted for alcohol use, to studies that adjusted for age at menopause, or to studies that adjusted for hormone therapy.

Conclusions: A higher education level may be associated with an increased risk of developing breast cancer, in which alcohol use, age at menopause, and hormone therapy may, at least partially, play a mediating role.

Key Words: Breast cancer – Cohort studies – Education – Incidence – Meta-analysis.

Unlike other diseases, a higher incidence of breast cancer has been shown among women with a higher socioeconomic status.¹ Cohort studies examining the association between education level, a widely used indicator of socioeconomic status, and breast cancer incidence, however, have yielded inconsistent results.²⁻¹⁹ Some reported an increased risk, whereas others did not detect an association. The inconsistent results from previous studies may be explained by differences in study populations, education level classifications, and statistical adjustments for confounding factors.

Women with a higher education level are likely to have advanced age at first birth and fewer children, both of which

put them at higher risk of developing breast cancer.^{20,21} In addition, a higher education level may be related to other factors that are associated with breast cancer. For example, a higher education level has been found to be associated with a later menopause onset,²² which is also a risk factor for breast cancer.²³ Therefore, these factors may play a mediating role in the association between education level and risk of breast cancer. Herein, we aimed to examine the association between education level and breast cancer incidence by conducting a meta-analysis of cohort studies, with consideration of possible mediating factors.

METHODS

Literature search

The present meta-analysis was conducted and reported according to the Meta-analysis of Observational Studies in the Epidemiology checklist.²⁴ We searched PubMed, Web of Science, and Scopus databases on June 10, 2018 for cohort studies that examined the association between education level and risk of breast cancer. An updated search was performed in December 2018. The following key words were used to detect potential studies: education, breast cancer, cohort study, prospective study, and follow-up study. We reviewed the reference lists of the retrieved articles as an additional search. Efforts were also made to contact authors of original studies for additional data if necessary.

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Study selection

Studies were eligible for analysis if they met the following criteria: the study design was a cohort study; the exposure of interest was education level; the endpoint of interest was breast cancer incidence; and risk estimates of breast cancer risk related to educational level with corresponding 95% CIs were provided. Case-control studies were not considered due to the higher risk of selection and recall bias in such studies.

Data extraction

The following information was extracted: family name of the first author, publication year, follow-up period, number of cases and participants, mean age or age range of the population, assessment of exposure and outcome, the age- and multivariate-adjusted risk estimates with corresponding 95% CI, and statistical adjustment for the potential confounding factors. Study quality was assessed using the Newcastle-Ottawa Quality Assessment Form for Cohort Studies.²⁵ Two authors independently performed the literature search, selection of studies, data extraction, and quality assessment, with disagreements resolved by discussion.

Statistical analysis

We used the relative risk (RR) as the common measure of the association. In the present meta-analysis, women with the lowest education level were treated as the reference group. For one study¹³ that reported results for subcohorts separately, the RRs of the subcohorts were combined and the summarized RR was then used for the meta-analysis. In one study in which both in situ and invasive breast cancers were reported, the RR for invasive breast cancer was used for the meta-analysis.¹¹ The I^2 statistics, the proportion of total variation contributed by between-study variation,^{26,27} were computed to measure the heterogeneity across studies. Because of the presence of heterogeneity in most analyses, a random-effects model²⁸ was used to compute the summary risk estimates throughout the study unless otherwise specified. We conducted a subgroup analysis by study area (Europe, the United States, or Asia) and reference group (education ≤ 9 or ≤ 12 y). We also performed a restricted analysis to studies with adjustment for possible mediating factors, including age at first birth, parity, and age at menopause. A sensitivity analysis was performed to examine the influence of an individual study on the pooled result by omitting one study in each turn. Potential publication bias was assessed by Begg's test and Egger's test.²⁹ All analyses were performed using STATA version 12.0 (StataCorp, College Station, TX).

RESULTS

Figure 1 shows the flow chart of study selection. After title and abstract screening, we obtained 56 studies for full-text review. Studies were excluded mainly because they were not cohort studies, risk estimates were not provided, or the outcome was not breast cancer incidence. Studies that

reported breast cancer mortality only were also excluded. Finally, we identified 18 cohort studies for meta-analysis, involving 194,654 incident cases and more than 10 million women.

Characteristics of the selected 18 cohort studies are presented in Table 1. The included studies were published between 1989 and 2015. Among the 18 studies, 10 were conducted in European countries, 5 were in the United States, and 3 in Asian countries. The number of cases diagnosed in the original studies ranged from 122 to 76,152, and the number of participants ranged from 1,716 to 4,553,484. Sources for education assessment varied among individual studies, with national statistics and self-reported questionnaires mostly used. Education >12 years (college or higher) was used as the highest education level group in all studies except for one study that used at least high school.¹⁶ Classification for the lowest education level group varied among individual studies, with education ≤ 9 years (basic education) mostly used. Fifteen studies used national cancer registers for breast cancer ascertainment, whereas the remaining three studies used medical records, self-reports, or death certificates. Four of the selected studies only adjusted for age, whereas the remaining studies adjusted for a wide range of potential confounding factors. The major adjusted covariates included age, age at first birth, parity, body mass index, alcohol use, age at menarche, family history of breast cancer, age at menopause, hormone therapy, and physical activity level (Supplementary Table 1, <http://links.lww.com/MENO/A480>). Quality scores of individual cohort studies using the Newcastle-Ottawa Quality Assessment Form ranged from 5 to 9 points, with a median of 6 points.

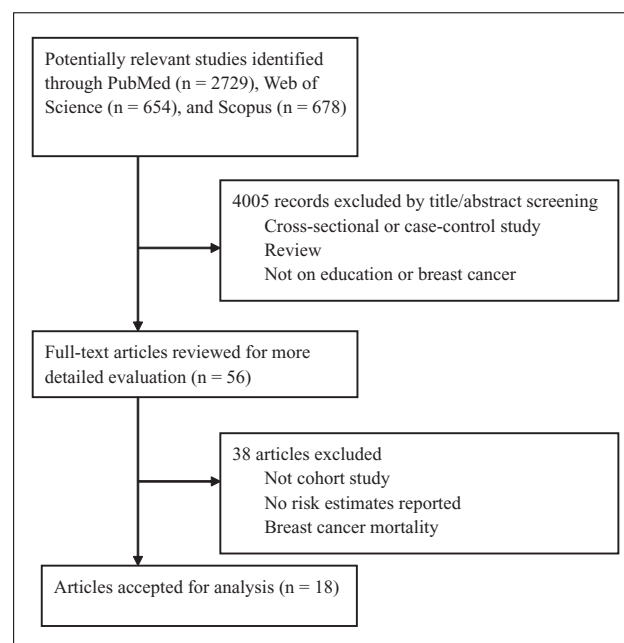


FIG. 1. The flow chart of study selection.

TABLE 1. Characteristics of cohort studies on the association between education level and breast cancer incidence

Study	Study area	Duration	Women (cases)	Exposure assessment	Outcome assessment	Highest vs lowest group	Quality score
Carter, 1989 ⁷	USA	1971-1984	7,426 (122)	Interview	Medical records or death certificates	More than college vs less than high school	9
Van Loon, 1994 ¹⁹	The Netherlands	1986-1989	1,716 (471)	Questionnaire	Cancer registries	University vs primary school	7
Heck, 1997 ¹²	USA	1971-1992	6,261 (229)	Interview	Medical records, self-reports, and death certificates	≥16 vs <12 y	8
Braaten, 2004 ¹⁴	Norway and Sweden	1991-1999	102,860 (1,090)	Questionnaire	Cancer registries	≥17 vs 7-9 y	6
Carlsen, 2008 ¹⁸	Denmark	1994-2003	1,590,000 (25,855)	Danish registers	Cancer registries	Higher education vs basic or high school	5
Fujino, 2008 ²	Japan	1988-2001	64,327 (169)	Questionnaire	Cancer registries	>12 vs <10 y	7
Mouw, 2008 ¹⁵	USA	1995-2003	195,674 (3,574)	Questionnaire	Cancer registries	Post graduate vs less than high school	6
Nishi, 2008 ⁹	Japan	1981-2003	20,136 (122)	Questionnaire	Cancer registries	≥13 vs ≤9 y	5
Hussain, 2008 ¹¹	Sweden	1990-2004	1,571,511 (43,222)	National census	Cancer registry	University vs <9 y	6
Spadea, 2009 ¹⁶	Italy	1985-1999	NA (7,771)	Population censuses	Cancer registry	At least high school vs primary school or less	6
Clegg, 2009 ³	USA	1973-2001	107,944 (1,739)	Self-report	Cancer registries	≥16 vs ≤11 y	6
Larsen, 2011 ⁵	Denmark	1993-2006	23,111 (907)	Statistics Denmark	Cancer registry	Higher education vs basic or high school	8
Menvielle, 2011 ⁴	Europe	1990-2006	242,095 (4,469)	Questionnaire	Cancer registries	University vs primary school	7
Beiki, 2012 ¹³	Sweden	1961-2007	4,553,484 (76,152)	Statistics Sweden	Cancer registry	≥13 vs ≤9 y	6
Palmer, 2012 ⁶	USA	1995-2009	59,000 (1,345)	Questionnaire	Self-report	≥17 vs ≤12 y	7
Meijer, 2013 ¹⁰	Denmark	2004-2008	1,539,162 (17,854)	Statistics Denmark	Cancer register	Tertiary vs basic education	5
Bjerkaas, 2015 ⁸	Norway	1974-2003	100,000 (7,490)	Official Norwegian Statistics	Cancer registry	≥13 vs <10 y	6
Goldberg, 2015 ¹⁷	Israel	1964-2008	40,586 (2,073)	Birth certificate	Cancer registries	>12 vs ≤8 y	7

The RRs of breast cancer incidence for each study and the overall combined RR are shown in Figure 2. The RRs ranged from 0.86 to 2.10 in individual studies, and 10 of them reached statistical significance. Compared with women in the lowest education level group, women in the highest education level group had a significantly increased risk of developing breast cancer (pooled RR = 1.22 [95% CI, 1.14-1.30]; $n = 18$). Considerable evidence of heterogeneity between studies was observed ($I^2 = 84.7\%$, P for heterogeneity <0.05).

We then performed a subgroup analysis by study area and reference group (Supplementary Table 2, <http://link-s.lww.com/MENO/A481>). The pooled RRs were 1.20 (95% CI, 1.11-1.30), 1.22 (95% CI, 1.05-1.21), and 1.35 (95% CI, 0.98-1.87) among studies conducted in Europe, the United States, and Asia, respectively. For studies using basic or high school (≤12 y) and using basic education (≤9 y) as the reference group, the pooled RRs were 1.22 (95% CI, 1.09-1.36) and 1.24 (95% CI, 1.12-1.38), respectively. Moderate-to-high heterogeneity remained in the subgroup analysis.

We next performed restricted analyses to studies with adjustment for potential mediating factors (Table 2). The association between education level and breast cancer incidence changed little when the results of 12 studies with control for age at first birth and parity (pooled RR = 1.22 [95% CI, 1.09-1.37]) were combined. The association was, however, attenuated and no longer significant when the analysis was restricted to nine studies that adjusted for alcohol use (pooled RR = 1.07 [95% CI, 0.99-1.14]; fixed-effect model), to five studies that adjusted for age at menopause (pooled RR = 1.10 [95% CI, 0.92-1.31]), or to five studies that adjusted for hormone therapy (pooled RR = 1.06 [95% CI, 0.98-1.14]; fixed-effect model).

The sensitivity analysis, which examined the influence of a single study on the results by omitting a study at each turn, yielded a range of RRs from 1.20 (95% CI, 1.13-1.27) to 1.24 (95% CI, 1.16-1.32), indicating that no single study seemed to materially alter the summary RRs. Neither Begg's test nor Egger's test provided evidence of publication bias (both $P > 0.80$).

DISCUSSION

The present study examined the association between education level and breast cancer incidence. Compared with women with a lower education level, women with a college or higher education level had a significantly increased risk of developing breast cancer. The association remained in studies with control for age at first birth and parity but was attenuated and no longer significant when the analysis was restricted to studies that adjusted for alcohol use, to studies that adjusted for age at menopause, or to studies that adjusted for hormone therapy.

The mechanisms by which higher education level was associated with increased risk of breast cancer remained unclear. One explanation may be that women with a higher education level were more likely to give birth to their first child at advanced age and have fewer children, both of which

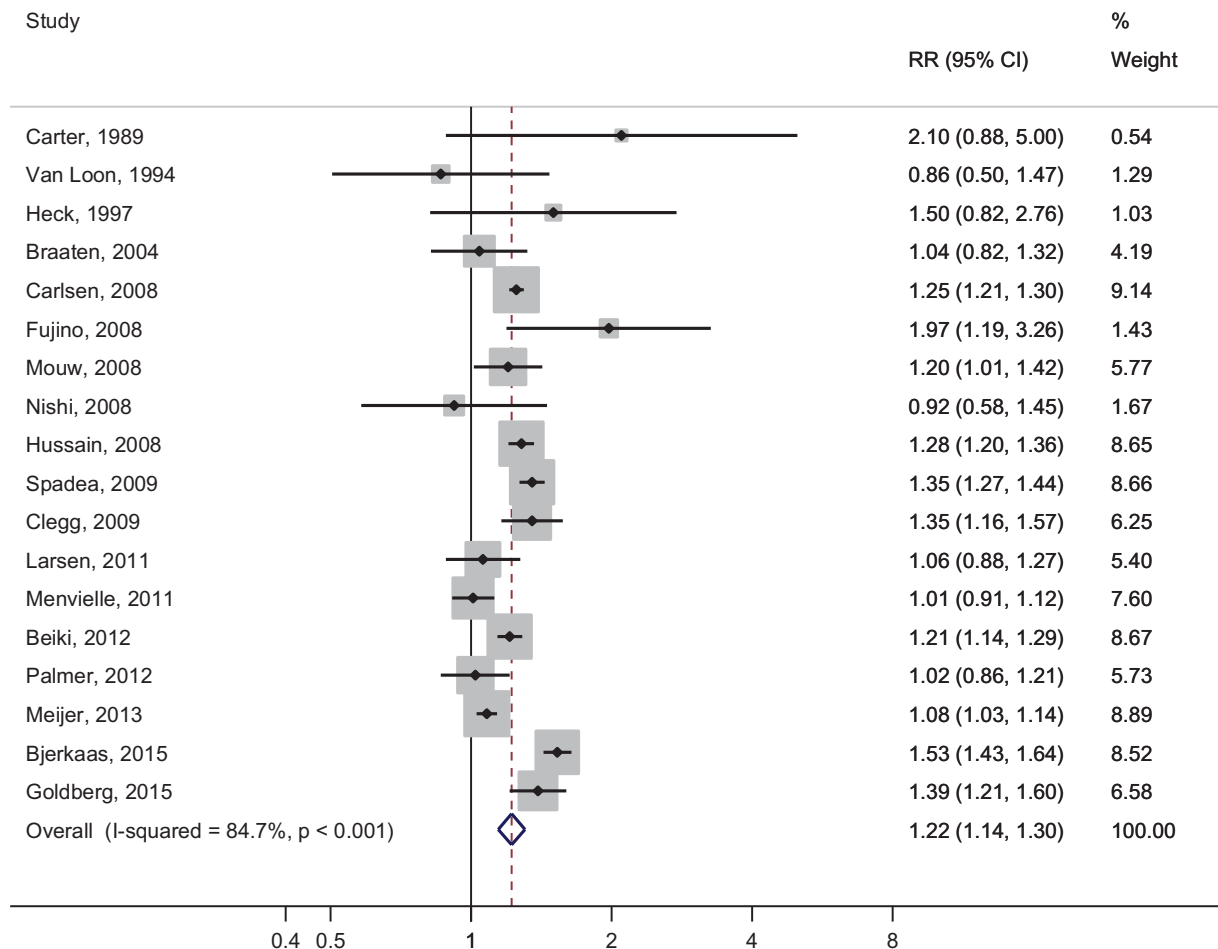


FIG. 2. Forest plot for the association between education level and breast cancer incidence. RR, relative risk; CI, confidence interval.

are associated with an increased risk of breast cancer.^{20,21} A significant association, however, persisted even after our restricted analysis to 12 studies with control for age at first birth and parity, indicating these two factors alone could not explain the observed increased risk. Another explanation may be related to other possible mediators, including alcohol use, age at menopause, and hormone therapy. Taking age at menopause as an example, a higher education level has been

found to be associated with a later menopause onset,²² which is also a risk factor for breast cancer.²³ In individual studies included in this meta-analysis, a higher education level was shown to be correlated with higher alcohol use,^{2,4,5,7,12,14,15,19} a later menopause,^{2,4,12,19} and higher prevalence of hormone therapy.^{5,12,15} The association disappeared in the restricted analysis to studies with control for alcohol use, age at menopause, or hormone therapy, indicating that the association

TABLE 2. Restricted analysis to studies with adjustment for possible mediators

Adjusted possible mediators	No. of studies	Multivariate-adjusted RR (95% CI)	I ² , %	Synchronously adjusted factors (no. of studies)	Age-adjusted RR (95% CI)
Age at first birth and parity	12	1.22 (1.09-1.37)	84	Alcohol use (9), age at menopause (5), HT (5)	1.37 (1.21-1.56)
Alcohol use	9	1.07 (0.99-1.14)	39	Age at first birth and parity (9), age at menopause (5), HT (5)	1.30 (1.16-1.46)
Age at menopause	5	1.10 (0.92-1.31)	52	Age at first birth and parity (5), alcohol use (5), HT (3)	1.25 (1.05-1.49)
HT	5	1.06 (0.98-1.15)	8	Age at first birth and parity (5), alcohol use (5), age at menopause (3)	1.25 (1.13-1.38)
At least three of the above	7	1.10 (0.98-1.23)	44	Age at first birth and parity (7), alcohol use (7), age at menopause (5), HT (5)	1.26 (1.12-1.40)

CI, confidence interval; HT, hormone therapy; RR, relative risk.

may be mediated, at least partially, by these factors. The results should, however, be treated with caution because the restricted analysis was based on a limited number of studies and the influences of other synchronously adjusted factors could not be excluded.

Furthermore, breast cancer screening adherence remained an alternative explanation for the association. Higher education levels were probably associated with higher screening adherence, which has been known to decrease breast cancer mortality but increase the incidence.^{30,31} Few studies, however, took screening adherence into account, and it is uncertain whether screening adherence has an effect on the association between education level and breast cancer incidence.

We observed considerable evidence of heterogeneity among individual studies, which may be explained by differences in population characteristics (eg, age and ethnic), education level classifications, duration of follow-up, and adjustments for covariates. Results from the restricted analysis suggested that discrepancies in adjustment for covariates may explain some of the observed heterogeneity. For example, there was no evidence of heterogeneity in studies that adjusted for alcohol use and in studies that adjusted for hormone therapy.

There were seven cohort studies that examined the association between education level and risk of breast cancer mortality.^{9,32-37} It seemed that the association of breast cancer mortality with education level was weaker than that of breast cancer incidence. Only one study³² reported a significant increased risk of breast cancer mortality associated with higher education level (RR = 1.14 [95% CI, 1.05-1.23]), and the others did not detect an association. It is difficult to explain mortality data because they involved both incidence and survival, which seemed to be influenced differently by education. In fact, in a previous meta-analysis conducted in Europe, higher socioeconomic status, measured by education, income, or occupation, was associated with a lower risk of case fatality, which may be partly explained by stage at diagnosis, treatment factors, comorbidity, and lifestyle factors.¹ Notably, adjustment for covariates was insufficient in the seven cohort studies on breast cancer mortality, with three only controlling for age and two only controlling for age, age at first birth, and parity. Given these reasons, we considered it inappropriate to combine the results of breast cancer mortality for a meta-analysis.

The main strengths of our meta-analysis included the large sample of participants involved and the cohort design of included studies, which minimized the likelihood of recall and selection biases. Limitations of the present study also merit discussion. First, as mentioned above, there was considerable evidence of heterogeneity across included studies, making the results of the meta-analysis hard to interpret. Second, control for potential confounding factors was not sufficiently considered in some studies. Several studies^{3,13,16,18} only adjusted for age, and the results may be confounded by other risk factors related to education level. Third, misclassification bias may have influenced the results. The risk of misclassification of breast cancer incidence would be low as breast cancer

ascertainment was based on national cancer registries in most studies. Misclassification of education level was, however, possible as about half of the included studies used self-reported data. Furthermore, whether the self-reported data were validated was seldom mentioned. Fourth, publication bias could affect the results of any meta-analyses. Little evidence of such bias was, however, observed in this meta-analysis, as indicated by Begg's and Egger's tests.

CONCLUSIONS

In summary, a higher education level may be associated with an increased risk of developing breast cancer, in which alcohol use, age at menopause, and hormone therapy may, at least partially, play a mediating role.

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